



Modelling Financial Markets

Chapter 2: Market Efficiency — From EMH to Adaptive Markets



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AMH: Adaptive Markets Hypothesis
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ BDS: Brock–Dechert–Scheinkman (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BIST: Borsa İstanbul (BIST 100 index) (the Istanbul Stock Exchange (BIST 100 index))
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BUX: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CD: Chow–Denning (multiple variance-ratio test)
- ▣ CI: Confidence Interval
- ▣ CME: Chicago Mercantile Exchange
- ▣ CRSP: Center for Research in Security Prices
- ▣ DFA: Detrended Fluctuation Analysis
- ▣ EMH: Efficient Market Hypothesis
- ▣ ETF: Exchange-Traded Fund
- ▣ FRED: Federal Reserve Economic Data
- ▣ FTSE: Financial Times Stock Exchange (FTSE Russell, index provider)
- ▣ FWER: Family-Wise Error Rate
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GMM: Generalized Method of Moments
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HK: Hong Kong

Acronyms used in this material (2/2)

- ▣ IPC: Índice de Precios y Cotizaciones (the Mexican Stock Exchange index)
- ▣ MDE: Minimum Detectable Effect
- ▣ MDH: Martingale Difference Hypothesis
- ▣ MIT: Massachusetts Institute of Technology
- ▣ ML: Machine Learning
- ▣ MSPE: Mean Squared Prediction Error
- ▣ NYSE: New York Stock Exchange
- ▣ OLS: Ordinary Least Squares
- ▣ PX: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ RW1: Random Walk 1 (i.i.d. increments)
- ▣ RW2: Random Walk 2 (independent increments)
- ▣ RW3: Random Walk 3 (uncorrelated increments)
- ▣ SE: Standard Error
- ▣ SPA: Superior Predictive Ability (test)
- ▣ TOM: Turn Of the Month
- ▣ TSMOM: Time-Series MOMentum
- ▣ US: United States
- ▣ VR: Variance Ratio
- ▣ WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)



Chapter Route and Outcomes

The question of this chapter

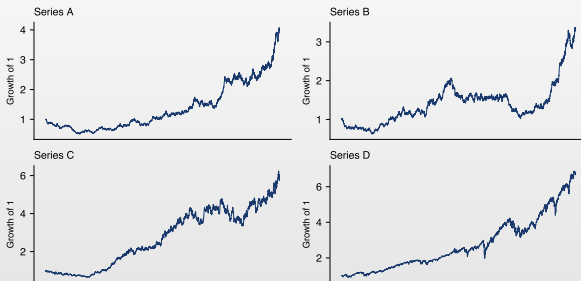
- If past prices revealed a strategy with positive expected risk-adjusted profit after costs, traders would exploit it until it disappeared
- A market in which this has already happened is called **efficient**; the claim restricts the statistical behaviour of returns, so it can be tested
- Part I: definitions, weak-form tests, predictive regressions, long memory
- Part II: adaptive markets, event studies, anomalies, behavioural finance, machine learning, a case study and one pass of the AI-assisted research loop; technical detail in the [Appendix](#)

After this chapter you can

- state which information set a test uses and which null it actually tests (RW1, RW3, martingale difference)
- run autocorrelation, variance-ratio, long-memory and calendar tests with inference robust to volatility clustering and to multiple testing
- separate a statistical rejection from an exploitable profit and from an irrational market

A Game: Which Series Is the Real Market?

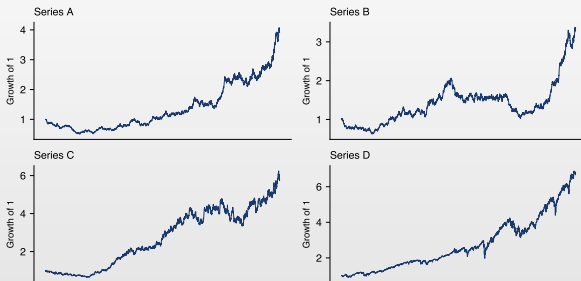
One of these is the S&P 500 (2010–2026); three are simulated random walks with the same drift and volatility



☐ One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?

A Game: Which Series Is the Real Market?

One of these is the S&P 500 (2010–2026); three are simulated random walks with the same drift and volatility



- One panel is the S&P 500 (2010–2026); three are random walks with the same mean and volatility. Vote: A, B, C or D?
- Answer: **D. Roberts (1959)**: random walks produce “patterns” by chance; we need statistics, not eyes
- **Bachelier (1900)** and **Samuelson (1965)**: properly anticipated prices fluctuate randomly, so unpredictability is the mark of a well-functioning market

Data Used in This Chapter

- 16 daily series up to 18 September 2026 (full list and sample starts: [▶ Appendix](#))
 - ▶ 13 equity indices from 2000: 4 developed (S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng), 9 emerging or frontier (BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai)
 - ▶ Bitcoin (from 2014; from 2011 in the yearly crypto analysis), Ethereum (from 2016) and EUR/RON, BNR reference rate (from July 2005)
- Daily log returns $r_t = \ln P_t - \ln P_{t-1}$, each series on its own calendar
- Equity indices: weekdays only; a close equal to the previous close is removed, since it is usually a holiday filled with the previous price
 - ▶ the rule also removes the rare genuine zero-return day; the effect on the test statistics is shown in the [▶ Appendix](#)
- Crypto-assets trade 7 days a week: 365 observations per year
- Analyses that combine several series (e.g. event studies) align prices on common trading days before computing returns

Efficiency and the Information Set

- Fama (1970): a market is **efficient** if prices “fully reflect” all available information $\mathcal{I}_t$
 - ▶ formally: no strategy using $\mathcal{I}_t$ earns returns above the fair (risk-adjusted) return

Form	Information set $\mathcal{I}_t$	Typical test (this chapter)
Weak	Past prices and returns	Autocorrelation, runs, variance ratios, long memory
Semi-strong	All public information (news, reports)	Event studies
Strong	Public and private (insider) information	Returns of insiders and fund managers

- Nested: strong $\Rightarrow$ semi-strong $\Rightarrow$ weak; risk-adjusted returns with factor models follow in [Chapter 3](#)

Efficiency as a Fair Game: the Martingale

- **Fair game** for excess returns: $\mathbb{E}[r_{t+1} - \mu_t \mid \mathcal{I}_t] = 0$, with μ_t the required (equilibrium) return
 - ▶ With a constant required simple return m , the total-return price discounted at m , $P_t/(1+m)^t$, is a martingale [Campbell, Lo & MacKinlay \(1997\)](#); discounting at the risk-free rate gives a martingale only under the risk-neutral measure
 - ▶ P_t itself is one only if $m = 0$: a zero expected **simple** return, not a zero expected log return
- “ $\mathbb{E}[P_{t+1} \mid \mathcal{I}_t] = P_t$ ” and “ $\mathbb{E}[\ln(P_{t+1}/P_t) \mid \mathcal{I}_t] = \mu$ ” are different restrictions
 - ▶ They differ by discounting and by the Jensen term: $\mathbb{E}[\ln(1+R)] \approx \mathbb{E}[R] - \text{Var}(R)/2$ for small returns ($\mathbb{E}[R]^2$ neglected)
- Consequence: $\text{corr}(\varepsilon_{t+1}, g(\mathcal{I}_t)) = 0$ for any function g of the past, $\varepsilon_{t+1} = r_{t+1} - \mu_t$
 - ▶ In particular all autocorrelations $\rho_k = \text{corr}(r_t, r_{t-k})$, $k \geq 1$, are zero when $\mu_t = \mu$
- Efficiency does not say prices are “correct” at every moment, only that pricing errors are not predictable; it says nothing about the variance: volatility may be predictable ([Chapter 1](#))

Random Walk Hypotheses RW1, RW2, RW3

- Log price $p_t = \ln P_t$ with $p_t = \mu + p_{t-1} + \varepsilon_t$; the three versions differ in what they assume about ε_t [Campbell, Lo & MacKinlay \(1997\)](#)
- **RW1:** ε_t i.i.d. (the strongest; rules out volatility clustering)
 - ▶ Rejected by the stylised facts of [Chapter 1](#)
- **RW2:** ε_t independent but not identically distributed (variance may change)
- **RW3:** ε_t uncorrelated: $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0$, while $\text{Cov}(\varepsilon_t^2, \varepsilon_{t-k}^2) \neq 0$ is allowed
 - ▶ The relevant hypothesis for real returns: tests must be robust to heteroskedasticity

The Joint-Hypothesis Problem

- ▣ “Fair return” requires a model of expected returns (e.g. CAPM, [Chapter 3](#))
- ▣ Every test of efficiency is a **joint** test of efficiency and of the asset-pricing model [Fama \(1991\)](#)
 - ▶ A rejection may mean an inefficient market **or** a wrong model of risk
 - ▶ Predictable **expected** returns are allowed if they compensate time-varying risk
- ▣ Some investors beat the market by luck: the question is whether success persists
- ▣ At daily horizons expected returns are tiny, so the choice of model matters little: weak-form tests are the cleanest
- ▣ Efficiency can never be “proved”, only not rejected by a given test on given data

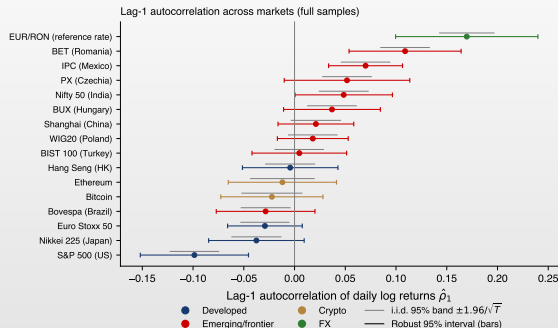
The Grossman–Stiglitz Paradox

- Gathering information is costly (analysts, data, computing)
- If prices reflected **all** information, nobody would be paid for collecting it [Grossman & Stiglitz \(1980\)](#)
 - ▶ Then nobody would collect it, and prices could not reflect it
- Resolution: an **equilibrium degree of inefficiency**, just large enough to reward informed traders for their costs
- Markets with cheap information and low costs (large, liquid) should be closer to efficiency: a prediction for our comparison of developed, emerging and crypto markets
- Nor does efficiency rule out bubbles: it says they are hard to time ([Chapter 17](#))

Autocorrelation and Its Robust Standard Error

- ▣ Sample autocorrelation: $\hat{\rho}_k = \sum_{t=k+1}^T e_t e_{t-k} / \sum_{t=1}^T e_t^2$, with $e_t = r_t - \bar{r}$
- ▣ Under RW1 (i.i.d.): $SE(\hat{\rho}_k) \approx 1/\sqrt{T}$; 95% band $\pm 1.96/\sqrt{T}$
- ▣ Unpredictable mean with volatility clustering (finite fourth moments): $SE(\hat{\rho}_k) \approx \sqrt{\hat{\tau}_k}$
 - ▶ $\hat{\tau}_k = \sum_t e_t^2 e_{t-k}^2 / (\sum_t e_t^2)^2$ (robust variance)
 - ▶ When large moves cluster, $\hat{\tau}_k > 1/T$: the i.i.d. band is too narrow and classical tests reject too often; every test below uses a variance that allows for this
- ▣ S&P 500: i.i.d. $SE = 0.012$, robust $SE = 0.027$, more than twice as large

Lag-1 Autocorrelation Across Markets



- Dots: $\hat{\rho}_1$ with robust 95% intervals; thin segments: the naive i.i.d. band
- Positive and significant: BET 0.109 (SE 0.028), IPC 0.070, EUR/RON 0.170
- Negative and significant: S&P 500 -0.099
- Bitcoin -0.022 , Ethereum -0.012 : not significant
- Candidate explanations: thin trading (BET), a managed exchange rate (EUR/RON), short-term overreaction and liquidity provision (S&P 500)
- Economic size: $R^2 = \hat{\rho}_1^2$, i.e. 0.97% (S&P 500) and 1.19% (BET) of the daily variance

Portmanteau Tests: Ljung–Box and Its Robust Version

- ▣ Ljung & Box (1978): $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi_m^2$ under i.i.d. returns
- ▣ Robust version: $\tilde{Q}(m) = \sum_{k=1}^m \hat{\rho}_k^2 / \hat{\tau}_k \rightarrow \chi_m^2$ under RW3 when the $\hat{\rho}_k$ are asymptotically uncorrelated (condition on the slide “The Heteroskedasticity-Robust Statistic z^* below)
- ▣ $m = 10$, 5% critical value 18.31:
 - ▶ S&P 500: $Q = 96.3$ but $\tilde{Q} = 19.9$ ($p = 0.031$)
 - ▶ BET: $Q = 108.0$, $\tilde{Q} = 24.4$ ($p = 0.007$)
 - ▶ Bitcoin: $Q = 20.2$ ($p = 0.028$) but $\tilde{Q} = 11.9$ ($p = 0.294$)
- ▣ The robust test keeps the rejections for S&P 500 and BET, but not for Bitcoin

The Runs Test

- Nonparametric: uses only the **signs** of returns (zero returns dropped), so it is immune to heavy tails [Wald & Wolfowitz \(1940\)](#)
- A **run** is a maximal block of equal signs: ++|-|+++ has 3 runs; Seminar A3 counts them on 20 signs
- With n_1 positive and n_2 negative signs, $n = n_1 + n_2$, under independence:
 - ▶ $\mathbb{E}[R] = \frac{2n_1n_2}{n} + 1$, $\text{Var}[R] = \frac{2n_1n_2(2n_1n_2 - n)}{n^2(n-1)}$, $z = \frac{R - \mathbb{E}[R]}{\sqrt{\text{Var}[R]}} \approx N(0, 1)$
- Too **few** runs ($z < 0$): signs persist; too **many** ($z > 0$): sign reversals; the test assumes a constant probability of a positive day
- Three markets: S&P 500 $z = 4.34$ (more runs, as $\hat{p}_1 < 0$); BET $z = -6.05$ (fewer runs, as $\hat{p}_1 > 0$); Bitcoin $z = 2.68$ although $\hat{p}_1$ is not significant
 - ▶ signs and magnitudes carry different information; all markets: [Appendix](#)

The Variance Ratio: Idea

- Uncorrelated increments with constant variance: variance grows linearly with the horizon,
 $\text{Var}(r_t + \dots + r_{t-q+1}) = q \text{Var}(r_t)$
 - ▶ q : the horizon in days (e.g. $q = 5$ is one trading week)
 - ▶ With changing variances (RW2, RW3) the variance of the sum is the sum of the variances; the sample $\widehat{\text{VR}}(q)$ still tends to 1 under the conditions of [Lo & MacKinlay \(1988\)](#)
- **Variance ratio:** $\text{VR}(q) = \frac{\text{Var}(r_t + \dots + r_{t-q+1})}{q \text{Var}(r_t)}$, equal to 1 for any covariance-stationary uncorrelated series
- For a covariance-stationary series, a weighted sum of autocorrelations:
 - ▶ $\text{VR}(q) = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \rho_k$
- $\text{VR}(q) > 1$: momentum (positive autocorrelation); $\text{VR}(q) < 1$: mean reversion
- Worked check: S&P 500 $1 + \hat{\rho}_1 = 0.901$ vs estimated $\text{VR}(2) = 0.901$

The Lo–MacKinlay Test

- Lo & MacKinlay (1988): overlapping q -day sums, with bias corrections

$$\blacktriangleright \hat{\sigma}_a^2 = \frac{1}{T-1} \sum_t (r_t - \hat{\mu})^2, \quad \hat{\sigma}_c^2(q) = \frac{1}{m} \sum_{t=q}^T \left(\sum_{j=0}^{q-1} r_{t-j} - q\hat{\mu} \right)^2$$

$$\blacktriangleright m = q(T - q + 1)(1 - q/T); \quad \widehat{VR}(q) = \hat{\sigma}_c^2(q) / \hat{\sigma}_a^2$$

- Under RW1 (homoskedastic):

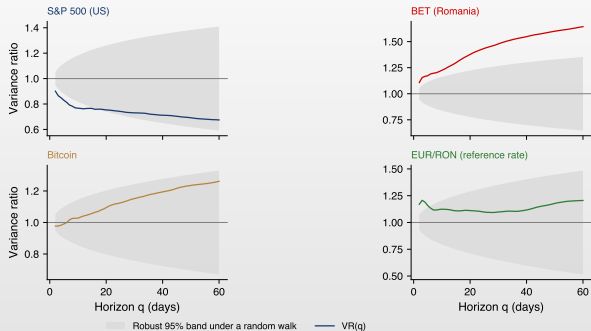
$$\blacktriangleright z(q) = \frac{\widehat{VR}(q) - 1}{\sqrt{2(2q-1)(q-1)/(3qT)}} \approx N(0, 1)$$

- Homoskedastic: constant variance; this z over-rejects when volatility clusters (next slide)

The Heteroskedasticity-Robust Statistic z^*

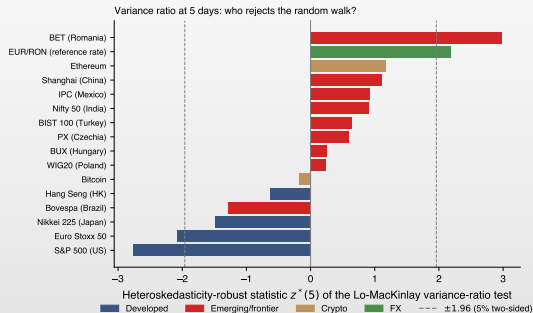
- Under RW3 plus the conditions of [Lo & MacKinlay \(1988\)](#) (finite moments, weak dependence (mixing), $\mathbb{E}[\varepsilon_t^2 \varepsilon_{t-j} \varepsilon_{t-k}] = 0$ for $j \neq k$), the $\hat{\rho}_k$ are asymptotically uncorrelated with variances $\hat{\delta}_k / T$
 - ▶ $\hat{\delta}_k = T \sum_{t=k+1}^T e_t^2 e_{t-k}^2 / (\sum_{t=1}^T e_t^2)^2$
- Asymptotic variance of $\widehat{VR}(q)$ and the test statistic:
 - ▶ $\hat{\theta}(q) = \sum_{k=1}^{q-1} \left(\frac{2(q-k)}{q} \right)^2 \hat{\delta}_k, \quad z^*(q) = \frac{\sqrt{T} (\widehat{VR}(q) - 1)}{\sqrt{\hat{\theta}(q)}} \approx N(0, 1)$
- RW3 alone is not enough: the conditions hold for GARCH with symmetric innovations, but can fail with a leverage effect
 - ▶ Without the last condition, $\hat{\theta}(q)$ needs the full covariance matrix of the $\hat{\rho}_k$
- S&P 500, $q = 2$: $\widehat{VR} = 0.901$, $z = -8.12$ but $z^* = -3.64$: still significant, but half as large in absolute value (Seminar B1 works it step by step)

Variance-Ratio Profiles



- $\widehat{VR}(q)$ for $q = 2, \dots, 60$; shaded: robust 95% band around 1, from $\hat{\theta}(q)$
- S&P 500: 0.901, 0.830, 0.753, 0.674 at $q = 2, 5, 20, 60$: reversal, outside the band at short horizons only
- BET: 1.108, 1.173, 1.377, 1.644: persistence accumulating with q
- Bitcoin: inside the band up to about one month; EUR/RON: short-run persistence of a managed float
- The band widens with q : a curve can point in one direction without being significant; direction is not evidence

Several Horizons and Markets: the Chow–Denning Test



- Bars: robust $z^*(5)$ for all 16 series; dashed lines ± 1.96
- Single horizon, reject at 5%: BET (2.98), EUR/RON (2.19), S&P 500 (−2.77), Euro Stoxx 50 (−2.08)
- Four horizons at 5% each: false-rejection probability up to $1 - 0.95^4 = 18.5\%$
 - **Chow & Denning (1993)**: $CD = \max_i |z^*(q_i)|$, $q = 2, 5, 10, 20$, 5% critical value 2.49
- Joint test rejects for 5 of 16: BET, S&P 500, IPC, EUR/RON, Ethereum; Bitcoin $p = 0.835$
- No ranking “developed = efficient”: the direction differs (reversal vs persistence)

Beyond Linear Predictability: the Martingale Difference Hypothesis

- ▣ **MDH:** $\mathbb{E}[r_t | \mathcal{I}_{t-1}] = \mu$: no function of the past predicts the mean, linear or not
- ▣ RW3 (all $\rho_k = 0$) is weaker. Counterexample: $r_t = \varepsilon_t + \theta(\varepsilon_{t-1}^2 - \sigma^2)$, ε_t i.i.d. symmetric
 - ▶ all $\rho_k = 0$, yet $\mathbb{E}[r_t | \varepsilon_{t-1}] = \theta(\varepsilon_{t-1}^2 - \sigma^2)$ is predictable: autocorrelation tests cannot see it
 - ▶ Tests of the MDH itself: the generalised spectral test [Escanciano & Velasco \(2006\)](#); an automatic portmanteau [Escanciano & Lobato \(2009\)](#) (details and results: [▶ Appendix](#))
- ▣ **BDS** [Brock, Dechert, Scheinkman & LeBaron \(1996\)](#) tests i.i.d., not the MDH: GARCH alone makes it reject
 - ▶ A BDS rejection is evidence against RW1, not against weak-form efficiency

Predictive Regressions and the Stambaugh Bias

- Model: $r_{t+1} = \alpha + \beta x_t + u_{t+1}$, $x_{t+1} = \theta + \rho x_t + v_{t+1}$, with $\text{Cov}(u, v) = \sigma_{uv}$
 - ▶ r : excess return; x : log dividend-price ratio $dp_t = \ln D_t - \ln P_t$, persistent ($\rho \approx 1$)
- x_t is predetermined, not strictly exogenous: OLS is consistent but biased [Stambaugh \(1999\)](#)
 - ▶ Project u on v : $u = (\sigma_{uv}/\sigma_v^2)v + e$, so $\hat{\beta} - \beta = (\sigma_{uv}/\sigma_v^2)(\hat{\rho} - \rho) + \text{noise}$
 - ▶ With $\mathbb{E}[\hat{\rho} - \rho] \approx -(1 + 3\rho)/T$: $\mathbb{E}[\hat{\beta} - \beta] \approx -\frac{\sigma_{uv}}{\sigma_v^2} \frac{1 + 3\rho}{T}$
- Price up $\Rightarrow dp$ down: $\sigma_{uv} < 0$, so the bias is **upward**: predictability looks stronger than it is
- Near a unit root the t -statistic is not $N(0, 1)$ either: valid inference by a Bonferroni Q -test over a confidence interval for ρ [Campbell & Yogo \(2006\)](#)

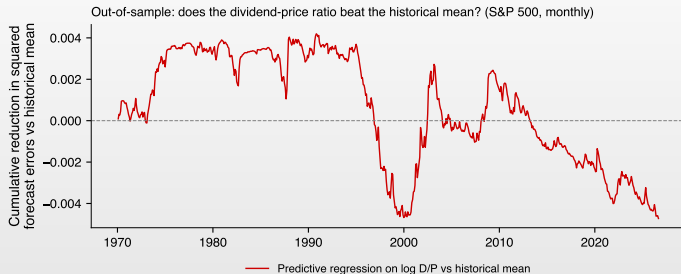
Worked Example: the Stambaugh Bias for the S&P 500

- ▣ Monthly log excess return on the lagged dp_t , 1934–2026, $T = 1\,111$ months (data construction: ▶ Appendix)
- ▣ OLS: $\hat{\beta} = 0.267$ (% per month per unit of dp), Newey–West $t = 0.91$: not significant even before any correction
- ▣ Inputs: $\hat{\rho} = 0.9988$, $\text{corr}(\hat{u}, \hat{v}) = -0.97$, $\hat{\phi} = \hat{\sigma}_{uv} / \hat{\sigma}_v^2 = -0.925$ (as fractions, not in %)
- ▣ Substitution: $\text{bias} = -100 \hat{\phi} (1 + 3\hat{\rho}) / T = 0.333$ (in %, like $\hat{\beta}$)
 - ▶ corrected slope $= 0.267 - 0.333 = -0.066$: the whole estimate is bias
 - ▶ the correction is itself estimated: formal inference by Campbell & Yogo (2006) or a bootstrap
- ▣ Simulation with the same ρ and $\text{corr}(u, v)$, $\beta = 0$, $T = 360$ months: the nominal 5% t -test rejects in 26.5% of samples

Out-of-Sample Predictability: R_{OS}^2 and the Clark–West Test

- Recursive forecasts: at each month t , re-estimate on data up to $t - 1$ only (expanding window) and forecast $\hat{r}_t$
 - ▶ benchmark: the historical mean $\bar{r}_t$ of the same data; $R_{OS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t (r_t - \bar{r}_t)^2$
 - ▶ Most predictors have $R_{OS}^2 < 0$ [Welch & Goyal \(2008\)](#); the 2022 update confirms it [Goyal, Welch & Zafirov \(2024\)](#)
 - ▶ Pseudo out-of-sample, as in [Welch & Goyal \(2008\)](#): monthly dividends are interpolated from quarterly totals, so dp_t is not strictly real-time
- Nested models: under H_0 the larger model estimates a zero β , which inflates its MSPE; the Diebold–Mariano test then rejects less often than its nominal level
 - ▶ [Clark & West \(2007\)](#): $f_t = (r_t - \bar{r}_t)^2 - [(r_t - \hat{r}_t)^2 - (\bar{r}_t - \hat{r}_t)^2]$; one-sided t on the mean of f_t
- [Campbell & Thompson \(2008\)](#): impose the sign of β and a non-negative forecast

Out-of-Sample Evidence: the Dividend–Price Ratio



- Cumulative squared-error gain over the historical mean, forecasts from 1970; rising line: dp helps
- 680 forecasts: $R_{OS}^2 = -0.51\%$; Campbell–Thompson restrictions bind in 5.7% of months, giving -0.51%
- Clark–West $t = 0.42$ (one-sided $p = 0.34$): no significant out-of-sample predictability
- The gain peaks in 1990 and is lost in the 1990s boom, when dp fell to record lows; after 2010 it erodes again
- Economic relevance: with monthly Sharpe ratio 0.160, even $R^2 = 0.5\%$ would raise by 20% the expected return of a mean–variance investor who times the market with the forecast (formula:

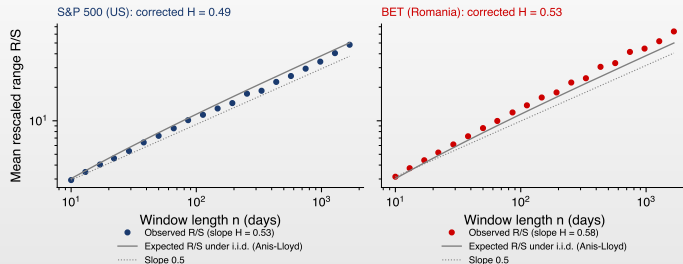
► Appendix)

 MFM_ch2_predictability

The Rescaled Range (R/S) Statistic

- **Hurst (1951)** found that wet years on the Nile followed wet years more often than chance allows, using the rescaled range (history: [Appendix](#))
- For a window of n returns with mean $\bar{r}_n$ and standard deviation s_n :
 - ▶ Cumulative deviations $Y_k = \sum_{i=1}^k (r_i - \bar{r}_n)$, $k = 1, \dots, n$
 - ▶ Range $R_n = \max_k Y_k - \min_k Y_k$; rescaled range R_n/s_n
- Scaling law: $\mathbb{E}[R_n/s_n] \approx c n^H$, where H is the **Hurst exponent**
 - ▶ Estimate: slope of $\ln(R/S)$ on $\ln n$, averaging R/S over non-overlapping windows
- $H = 0.5$: no long memory (short-run dependence allowed); $H > 0.5$: persistence; $H < 0.5$: anti-persistence
 - ▶ **Long memory**: autocorrelations decay so slowly that their sum diverges, $\rho_k \sim k^{2H-2}$

R/S Analysis: S&P 500 and BET



- Dots: average R/S by window length n ; line: the [Anis & Lloyd \(1976\)](#) expectation under i.i.d. Normal returns (exact formula: [Appendix](#))
- Small-sample bias: even for i.i.d. returns the raw slope exceeds 0.5; the Anis-Lloyd corrected exponent $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n]$
- S&P 500: raw 0.530, corrected 0.486: the “persistence” was the artefact
- BET: raw 0.578, corrected 0.533: persistence remains
- Open: is BET's $H > 0.5$ long memory or just $\hat{\rho}_1 = 0.109$? Classical R/S cannot tell

Hurst Exponent and Variance Ratio: One Horizon Is Not Enough

- ▣ For exactly self-similar increments (fractional Gaussian noise): $\text{Var}(r_t + \dots + r_{t-q+1}) = q^{2H} \text{Var}(r_t)$
- ▣ Hence $\text{VR}(q) = q^{2H-1}$ and the **implied** $H(q) = 0.5 + \ln \text{VR}(q) / (2 \ln q)$: BET 0.553, S&P 500 0.453 at $q = 20$
- ▣ Short-memory counterexample: an AR(1) with $\rho_1 = -0.1$, $\rho_k = (-0.1)^k$
 - ▶ $\text{VR}(20) = 0.826$, so $H(20) = 0.468$: “anti-persistent” without any long memory
 - ▶ yet $H(250) = 0.482$: for short memory $H(q) \rightarrow 0.5$ as $q \rightarrow \infty$ (Seminar A5–A6)
 - ▶ an AR(1) with BET’s $\hat{\rho}_1 = 0.109$ gives $H(20) = 0.535$: most of BET’s value is short memory
- ▣ Separating the two needs a statistic built for it: Lo’s modified R/S (next slide)

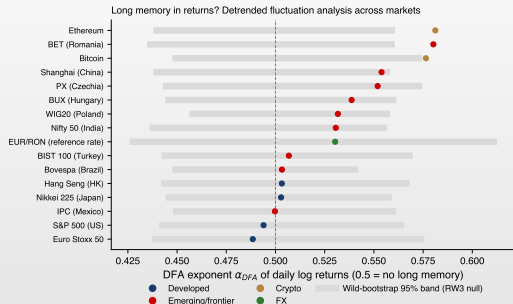
Lo's Modified R/S Test

- Lo (1991): short-run autocorrelation inflates R/S and mimics long memory
- Replace s_n by a long-run (HAC) standard deviation with ℓ lags:
 - ▶ $\hat{\sigma}_\ell^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^{\ell} \left(1 - \frac{j}{\ell+1}\right) \hat{\gamma}_j$, $\hat{\gamma}_j$: autocovariance at lag j
 - ▶ $V = \frac{\max_k Y_k - \min_k Y_k}{\hat{\sigma}_\ell \sqrt{T}}$; bandwidth ℓ chosen as in Andrews (1991)
- Under short memory, V lies in $[0.809, 1.862]$ with probability 95%
- Full samples: S&P 500 $V = 1.559$, Bitcoin 1.547, PX 1.750, EUR/RON 1.429: not rejected
 - ▶ BET $V = 1.961$ ($\ell = 7$) and Ethereum $V = 1.980$: just above 1.862, weak evidence of long memory

Detrended Fluctuation Analysis (DFA)

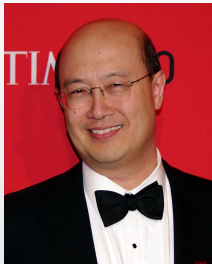
- Peng et al. (1994): linear detrending of the profile removes a locally constant mean of returns; trends in returns need higher-order DFA
- Algorithm (Seminar B3 shows one window step by step):
 - ▶ Profile $y_k = \sum_{i=1}^k (r_i - \bar{r})$; split into boxes of length n
 - ▶ In each box fit a linear trend and keep the residuals
 - ▶ $F(n)$ = root mean square of all residuals; repeat for many n
- $F(n) \propto n^{\alpha_{DFA}}$: the DFA exponent α_{DFA} plays the role of H ($\alpha_{DFA} = 0.5$: no long memory)
- Null band by **wild bootstrap**: $e_t \eta_t$, $\eta_t = \pm 1$ with probability 1/2 (199 replications)
 - ▶ Keeps the volatility path $|e_t|$ (allowed under RW3) and destroys serial correlation; shuffling would destroy both
 - ▶ Implicit null: signs symmetric and independent given $|e_t|$, stronger than RW3: a benchmark, not an exact RW3 test and not a confidence interval for α_{DFA}

DFA Exponent Across Markets



- Dots: α_{DFA} ; bars: sign-flip 95% null band (volatility kept, no autocorrelation)
- Inside the band: S&P 500 0.494, Euro Stoxx 50 0.488, Nikkei 0.503, PX 0.552, Shanghai 0.554
- Above (3 of 16): Ethereum 0.581, BET 0.580 (band to 0.561), Bitcoin 0.576, borderline (band to 0.575)
- A shuffled band flags 5: PX and Shanghai were “memory” only through volatility clustering
- Cross-checks: Lo’s V rejects BET and Ethereum marginally, not Bitcoin; the local Whittle estimator of the long-memory parameter $d = H - 0.5$ for BET: $\hat{d} = 0.118$, 95% CI [0.062, 0.174] ([Appendix](#))
- Lesson: memory in returns is weak and fragile; memory in volatility ([Chapter 1](#)) is strong

The Adaptive Market Hypothesis



Andrew W. Lo, MIT

Photo: David Shankbone (2012); CC BY 3.0;

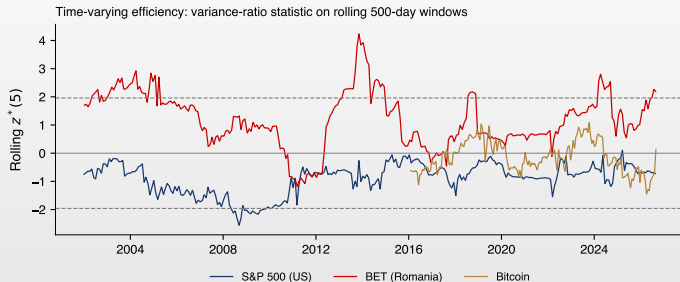
Wikimedia Commons

- Lo (2004); book-length version Lo (2017)
- Markets are an ecology of competing species (traders, funds, algorithms) that adapt by trial and error, not by perfect optimisation
 - ▶ efficiency is a degree that depends on the number of competitors, the profit opportunities and the environment
- Prediction: efficiency **varies over time and with market conditions**; anomalies appear, weaken and may reappear
- Caution: time-varying statistics alone do not identify the AMH, since an EMH with time-varying risk premia also allows them (comparison table:
▶ [Appendix](#))
- Evidence of time variation: survey [Lim & Brooks \(2011\)](#); long-run indices [Urquhart & McGroarty \(2016\)](#)

Measuring Time-Varying Efficiency

- Compute a statistic on a **rolling window**: 500 trading days (about two years), moved forward by 21 days (one month)
 - ▶ Statistics: $z^*(5)$ of the variance-ratio test and the DFA exponent
- Null band for DFA: wild bootstrap **inside each window** (199 sign flips), so the band follows the window's volatility
 - ▶ An i.i.d. Student- t_4 band $[0.378, 0.614]$ ignores volatility clustering, which RW3 allows and which raises α_{DFA}
- Reading the results
 - ▶ Overlapping windows are strongly correlated: runs of rejections are not independent evidence
 - ▶ Even under the null about 5% of windows reject; a share near 5% is a description, not a calibrated test of stability

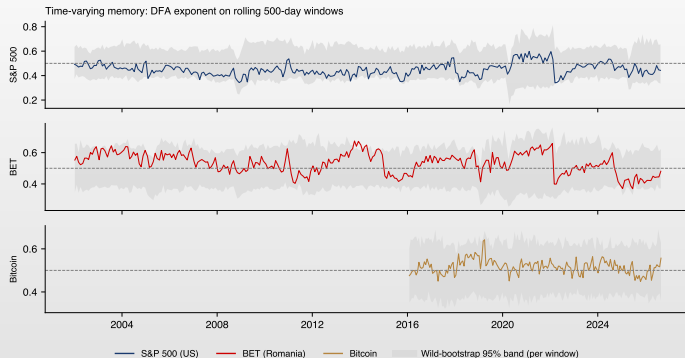
Rolling Variance Ratio: S&P 500, BET, Bitcoin



- $z^*(5)$ on rolling 500-day windows; dashed lines ± 1.96
- Share of windows with $|z^*(5)| > 1.96$: S&P 500 6.4%, BET 22.4%, Bitcoin 0.0%
- S&P 500: the full-sample rejection sits in a few turbulent windows
- BET: rejections cluster in episodes, then fade; FTSE upgrade to Secondary Emerging effective 21 Sep 2020, a candidate change in the investor base (Seminar C1)
- Bitcoin, 2014–2026: no window rejects

 [MFM_ch2_rolling_efficiency](#)

Rolling DFA Exponent: S&P 500, BET, Bitcoin

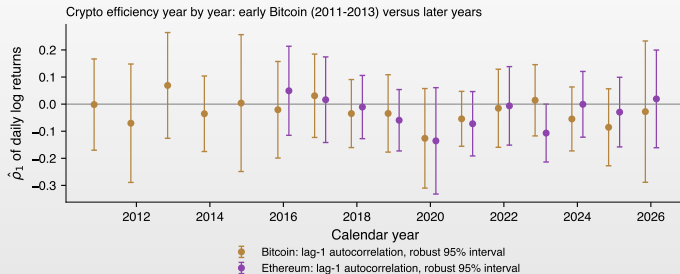


- α_{DFA} on rolling 500-day windows; shaded: wild-bootstrap 95% band computed in each window
- Outside their own band: S&P 500 2.0%, BET 2.7%, Bitcoin 0.0% of windows
- With the fixed i.i.d. Student- t_4 band: 6.4%, 12.9%, 1.1%: the apparent BET “episodes” were volatility clustering
- Low power: the BET band averages [0.36, 0.64]; 500 observations cannot pin down a scaling exponent

The AMH, Tested

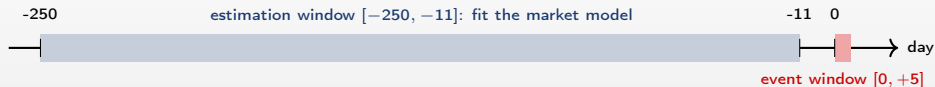
- Kim, Shamsuddin & Lim (2011): century-long US data; predictability varies with market conditions (crises, bubbles), not along a trend
 - ▶ This turns the AMH narrative into a regression with testable coefficients
- Our version: regress the rolling $z^*(5)$ on the window's annualised volatility (%) and a crisis dummy (window overlaps Sep 2008–Jun 2009 or Feb–Jun 2020)
 - ▶ Newey–West errors with 24 lags: consecutive windows share 95% of their days
- S&P 500: crisis -0.67 ($t = -2.37$), volatility 0.006 ($t = 0.27$), $R^2 = 0.20$
- BET: crisis -0.82 ($t = -2.77$), volatility -0.008 ($t = -0.36$), $R^2 = 0.15$
- Crisis windows are associated with lower $z^*(5)$ (towards reversal) in both markets, beyond volatility: a conditional association, not a causal effect
 - ▶ z^* mixes $\widehat{\sqrt{R}} - 1$ and its standard error; state dependence is consistent with the AMH but does not identify it uniquely

Crypto Efficiency Year by Year



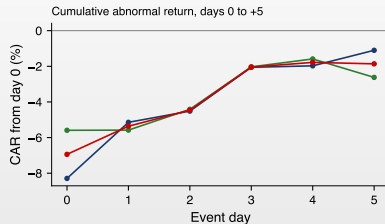
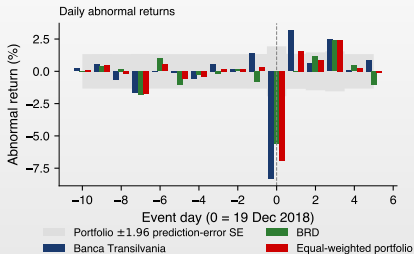
- Lag-1 autocorrelation for each calendar year, robust 95% intervals; Bitcoin from 2011
 - Only 1 of 27 yearly $\hat{\rho}_1$ has $|t| > 1.96$ (Ethereum 2023, -0.107), against $0.05 \times 27 = 1.35$ expected by chance
 - Unlike [Urquhart \(2016\)](#), who rejects efficiency in the early years: other tests (including BDS), another price source and classical errors (literature: [Appendix](#))
 - Power: one year of 365 returns gives a robust SE of about 0.06–0.13, so small effects are invisible
- [MFM_ch2_crypto_efficiency](#)

The Event-Study Method



- Fama, Fisher, Jensen & Roll (1969): how fast do prices absorb public news (stock splits)?
- Market model: $r_{i,t} = a_i + b_i r_{m,t} + \varepsilon_{i,t}$, $r_{m,t}$ the market return; abnormal return $AR_{i,t} = r_{i,t} - \hat{a}_i - \hat{b}_i r_{m,t}$;
 $CAR_i[\tau_1, \tau_2] = \sum_{t=\tau_1}^{\tau_2} AR_{i,t}$
- First-pass test: $t = CAR_i / (\hat{\sigma}_\varepsilon \sqrt{L})$, L the event-window length MacKinlay (1997)
- For unanticipated news, efficiency predicts a jump on day 0 and no predictable drift or reversal afterwards
- Daily data can only show absorption within a day; speed measured in microseconds needs intraday data (Chapter 10; literature: [Appendix](#))

Worked Example: the Romanian Bank Tax of December 2018



- Tax announced on the evening of 18 Dec 2018, day 0 = 19 Dec; Banca Transilvania (TLV), BRD and their equal-weighted portfolio vs BET (Seminar 0, B7)
- TLV: $\hat{b} = 1.18$, $\hat{\sigma}_\varepsilon = 0.83\%$, $AR_0 = -8.3\%$ ($t = -10.0$); $CAR[0, +5] = -1.1\%$ ($t = -0.54$), since $CAR[+1, +5] = +7.2\%$ ($t = 3.82$)
- Most of the day-0 fall reverses within the week: an **overreaction**, against semi-strong efficiency, or later news (the tax was softened); one event cannot separate them

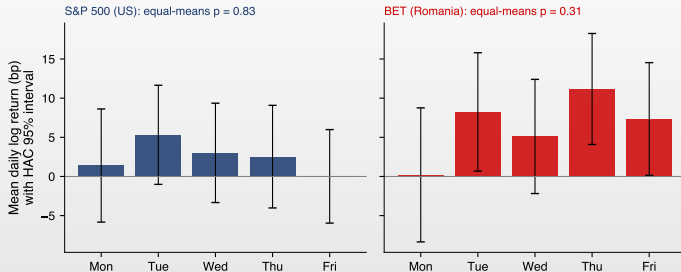
Event Studies: Inference Done Properly

- Prediction-error variance: $\text{Var}(AR_\tau) = \sigma_\varepsilon^2 \left[1 + \frac{1}{T_0} + \frac{(r_{m,\tau} - \hat{\mu}_m)^2}{(T_0 - 1)\hat{\sigma}_m^2} \right]$, T_0 = estimation days [MacKinlay \(1997\)](#)
 - ▶ Bank tax, BET $r_{m,0} = -11.9\%$: the bracket is 2.12, so TLV's t shrinks in absolute value from -10.0 to -6.85 ; BRD: $AR_0 = -5.6\%$, $t = -4.03$
- Several firms: standardise each AR by its own prediction error and use the cross-sectional variance, robust to event-induced variance [Boehmer, Masumeci & Poulsen \(1991\)](#)
- Same event day for all firms: abnormal returns are cross-correlated and cross-sectional tests over-reject; the equal-weighted portfolio test already contains the correlation
 - ▶ TLV and BRD: $\bar{r} = 0.15$; mean AR_0 with an independence $t = -7.55$, portfolio $t = -7.03$
 - ▶ [Kolari & Pynnönen \(2010\)](#) correct a cross-sectional test by $\sqrt{(1 - \bar{r})/(1 + (N - 1)\bar{r})}$; with ten such firms the test would understate the variance of the mean AR 2.81 times

Anomalies and Calendar Effects

- An **anomaly** is a return pattern the asset-pricing model cannot explain: mispriced risk, behavioural bias or data mining
 - ▶ Families: calendar, time-series (index momentum), cross-sectional (size, value, momentum: [Chapter 3](#))
 - ▶ Returns 26% lower out of sample and 58% lower after publication [McLean & Pontiff \(2016\)](#)
- Calendar effects: Monday [French \(1980\)](#); January [Rozeff & Kinney \(1976\)](#); turn of the month (TOM: last trading day and first three of the next month) [Ariel \(1987\)](#); [Lakonishok & Smidt \(1988\)](#)
 - ▶ Day of week: $r_t = \sum_{d=1}^5 \beta_d D_{d,t} + u_t$, Wald test of $\beta_1 = \dots = \beta_5$; TOM and January: one dummy each
 - ▶ Newey–West HAC errors [Newey & West \(1987\)](#), $L = 5$ days or 3 months: residuals are heteroskedastic and autocorrelated

Day-of-the-Week Effect: S&P 500 and BET



- Mean daily log return by weekday, in basis points (bp; 1 bp = 0.01%), 2000–2026, HAC 95% intervals
- S&P 500: from 0.0 bp (Friday) to 5.3 bp (Tuesday); equal-means $p = 0.83$: no Monday effect since 2000, unlike French (1980)
- BET: Thursday 11.2 bp ($t = 3.09$), yet the joint test gives $p = 0.31$
- Two different nulls: a day's mean equal to zero (mixes the day effect with the positive average return) vs all five means equal (the day effect itself)

 MFM_ch2_calendar_anomalies

January and Turn-of-the-Month Effects

Market	January extra (%/month)	t_{HAC}	TOM extra (bp/day)	t_{HAC}	p
S&P 500 (US)	-0.66	-0.80	3.2	0.93	0.3539
BET (Romania)	3.48	1.49	17.3	3.62	0.0003
Euro Stoxx 50	-0.30	-0.29	5.1	1.21	0.2266
WIG20 (Poland)	0.53	0.40	8.2	1.73	0.0841
BUX (Hungary)	2.66	2.25	15.1	3.63	0.0003

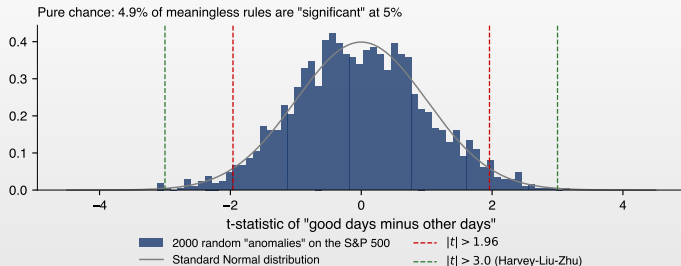
- TOM: BET +17.3 bp per day and BUX +15.1 bp, both with $|t| > 3.5$; S&P 500 only +3.2 bp
- January: large but imprecise for BET ($t = 1.49$); BUX, with $t = 2.25$, is the only market with a January-effect p -value below 0.05
- We ran 15 tests (5 markets $\times$ 3 effects): which survive a correction for multiple testing?

Multiple Testing: Bonferroni and Holm

- **FWER** (family-wise error rate): the probability of at least one false rejection; with M independent tests at level α , $\text{FWER} = 1 - (1 - \alpha)^M$
 - ▶ $M = 15$, $\alpha = 5\%$: $\text{FWER} = 53.7\%$, a coin flip
- **Bonferroni**: reject only if $p_i < \alpha/M$; guarantees $\text{FWER} \leq \alpha$ for any dependence
- **Holm**: sort $p_{(1)} \leq \dots \leq p_{(M)}$; reject while $p_{(j)} < \alpha/(M - j + 1)$, stop at the first failure
 - ▶ Same guarantee, never less powerful than Bonferroni
- Our 15 calendar tests: Bonferroni threshold 0.0033; 3 raw rejections, 2 survive Holm (BET and BUX turn of the month); Holm stops at BUX January
- Seminar B6 carries out the ranking step by step; bootstrap methods that use the dependence between tests:

▶ Appendix

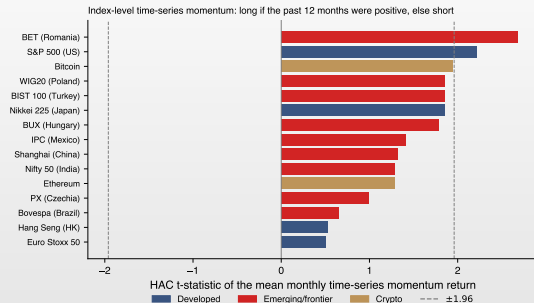
How Many “Anomalies” Does Pure Chance Produce?



- 2000 meaningless rules on the S&P 500: each labels a random 20% of days as “good” and tests good minus other days
- 4.9% have $|t| > 1.96$, the 5% the test promises; the largest $|t|$ is 3.19
- Whoever tries many rules and reports the best is very likely to find a “significant anomaly”; data snooping in trading rules: [Sullivan, Timmermann & White \(1999\)](#)
- [Harvey, Liu & Zhu \(2016\)](#) suggest $|t| > 3.0$ for a new factor; here 0.25% of random rules pass. A higher bar is a heuristic, not a substitute for defining the family of tests tried

 [MFM_ch2_multiple_testing](#)

Momentum and Index-Level Time-Series Momentum



- Reversal of 3–5-year losers De Bondt & Thaler (1985); cross-sectional momentum over 3–12 months Jegadeesh & Titman (1993) (Chapter 3)
- TSMOM Moskowitz, Ooi & Pedersen (2012): signal $s_t = \text{sign}(\sum_{j=1}^{12} r_{t-j})$ of past monthly log returns; strategy return $s_t R_t$; bars: HAC t (6 lags)
- Significant: BET ($t = 2.68$, 1.40% a month), S&P 500 ($t = 2.22$); Bitcoin $t = 1.94$
- Sharpe ratio below buy-and-hold: BET 0.65 vs 0.75, S&P 500 0.48 vs 0.52: much of TSMOM is the equity premium earned while long
- Data limits: price indices without financing rates or dividends, 15 markets tested; full table: [Appendix](#)

 MFM_ch2_tsmom

Psychology in Prices, from the Floor to the Feed



New York Stock Exchange trading floor, 1963

Photo: Thomas J. O'Halloran (1963); public domain;
Wikimedia Commons

- **Excess volatility** Shiller (1981): prices move far more than the dividends that should justify them; **overreaction** De Bondt & Thaler (1985): extreme losers become winners over 3–5 years
- **Prospect theory** Kahneman & Tversky (1979): losses loom larger than gains relative to a reference point
 - ▶ Disposition effect: winners sold too early, losers held too long
- **Overconfidence**: too much trading; **anchoring** and **limited attention**: slow reaction to news, a candidate cause of momentum
- **Herding** Pele, Mazurencu-Marinescu & Nijkamp (2013): once voices on a floor, now order books, algorithms and social media (Chapter 10)
- Biases alone do not break efficiency: rational arbitrageurs could trade against them. Can they?

Limits to Arbitrage



Andrei Shleifer, Harvard

Photo: Manuel Castells (2018); CC BY-SA 2.0;

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- **Shleifer & Vishny (1997)**: real arbitrage is risky and needs capital
- **Noise-trader risk**: mispricing can get worse before it corrects
 - ▶ Noise traders: investors trading on non-information (sentiment, rumours)
- **Funding constraints**: managers lose clients after losses and must close positions at the worst time
- **Costs**: short-selling limits, margins, trading costs, no perfect substitutes
- **Anomalies can persist where arbitrage is hard**: in small, illiquid markets such as BET we indeed find the strongest persistence
- **The AMH reconciles both views**: rational and biased behaviour coexist and evolve

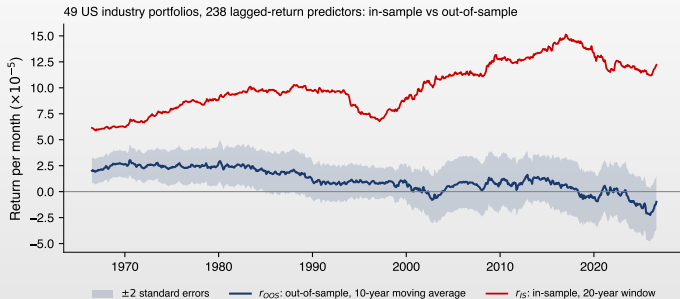
Can Machine Learning Beat Efficient Markets?

- ▣ ML (machine learning) learns a prediction rule from data and captures nonlinear patterns that autocorrelation tests miss ([Chapter 13](#))
- ▣ [Gu, Kelly & Xiu \(2020\)](#): trees and neural networks give large economic gains over linear regressions in the US cross-section, yet their monthly out-of-sample R^2 is below 1%
- ▣ Large language models read news and filings: [Tak & Pele \(2025\)](#) forecast stock trends with LLaMA
- ▣ What ML adds to the cautions of this chapter: **chronological validation**
 - ▶ train only on data available before each forecast (walk-forward); random cross-validation leaks the future (look-ahead bias)
- ▣ Key question: when many predictors are available, what does an in-sample rejection of efficiency mean? The case study answers it

Case Study: Rational Learning with Many Predictors

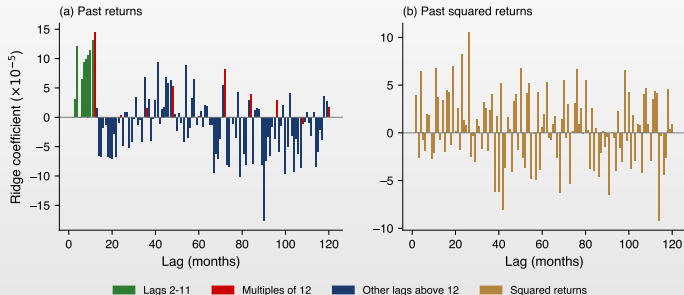
- **Martin & Nagel (2022)**: investors price N assets whose cash flows depend on J characteristics with **unknown** coefficients and learn them optimally (Bayes) from past data
 - ▶ the posterior mean is a ridge regression: coefficients shrunk towards 0; investors are risk-neutral, so no predictability can be a risk premium
- **In sample**: with $J/N \rightarrow \psi > 0$ the test of no predictability rejects with probability tending to 1 (**Martin & Nagel (2022)**, Prop. 4)
 - ▶ a rejection is not evidence of irrationality: the rational-expectations null is the wrong benchmark when parameters must be learned
- **Out of sample**: a portfolio built with coefficients from other periods has $E[r_{oos}] = 0$ (**Martin & Nagel (2022)**, Prop. 6)
 - ▶ out-of-sample tests keep their economic meaning: they are the right test of efficiency
- US stocks: in-sample R^2 about 1%, stable; out of sample much lower and within two standard errors of 0 after 2014 (**Martin & Nagel (2022)**, Fig. 8). Model, simulations and design: [▶ Appendix](#)

Replication on Course Data: In Sample vs Out of Sample



- 49 industry portfolios ([Kenneth French Data Library](#)), July 1926 – August 2026, 238 lagged-return predictors, rolling 20-year OLS, specification of [Martin & Nagel \(2022\)](#), Sec. 7
- r_{IS} (in-sample return of the predicted-return portfolio): 5.90 to 15.14 ($\times 10^{-5}$ a month), positive by construction; mean in-sample $R^2 = 5.4\%$
- r_{OOS} (out-of-sample return of the same portfolio), 842 months: mean 1.05×10^{-5} , about 11% of r_{IS} ; to June 1986: 4.16 SE above 0, afterwards 0.62
- Bands (± 2 SE) are descriptive, not a test of $E[r_{OOS}] = 0$; 49 portfolios instead of thousands of stocks make J/N large

Replication on Course Data: Ridge Coefficients by Lag



- Full-sample ridge, January 1971 – August 2026, penalty by leave-one-year-out cross-validation (56 years); compare [Martin & Nagel \(2022\)](#), Fig. 7
- Lags 2–11: 8 of 10 positive, **momentum** [Jegadeesh & Titman \(1993\)](#); multiples of 12: 9 of 10 positive, **seasonality** [Heston & Sadka \(2008\)](#)
- Other lags above 12: 67% negative, long-run **reversal** [De Bondt & Thaler \(1985\)](#); squared returns: no dominant sign
- Cross-validated $R^2 = 0.013\%$ (OLS: -2.26%): shrinkage recovers the known anomalies, yet they predict almost nothing in the years left out

 MFM_ch2_big_data

Hypothesis: Did Spot ETFs Make Bitcoin More Efficient?

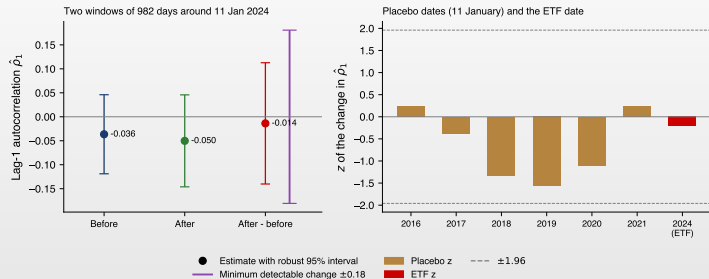
- A typical assistant proposal: “US spot Bitcoin ETFs (trading since 11 January 2024) brought arbitrageurs, so Bitcoin became more efficient”
- Discovery loop in one pass (reusable prompts: [▶ Appendix](#))
 - ▶ **literature:** [Kia, Liu, Li, Song & Xu \(2025\)](#) find that after the spot ETFs the spot market became the dominant venue for price discovery again; [Tran & Leirvik \(2020\)](#): crypto efficiency varies and has increased; resolve every DOI yourself
 - ▶ **hypothesis:** $H_0 : \rho_{1,\text{post}} = \rho_{1,\text{pre}}$, two windows of equal length around the regulatory date
 - ▶ **data and code:** daily Bitcoin closing price (USD); first replicate the chapter's full-sample $\hat{\rho}_1 = -0.022$
- Checks fixed before seeing results
 - ▶ robust standard errors (not $1/\sqrt{T}$); equal windows; the regulatory date only, plus placebo dates; a control asset for confounders (the April 2024 halving, rates, inflows); a correction for every statistic tried; the minimum detectable effect

Estimation and Power

- Test: two windows of 982 days around 11 January 2024; $z = \Delta\hat{\rho}_1 / \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$, robust standard errors s
 - ▶ before (May 2021 – Jan 2024): $\hat{\rho}_1 = -0.036$ ($s = 0.042$), $\text{VR}(5) = 1.000$ ($z^* = 0.00$)
 - ▶ after (Jan 2024 – Sep 2026): $\hat{\rho}_1 = -0.050$ ($s = 0.049$), $\text{VR}(5) = 0.939$ ($z^* = -0.64$)
 - ▶ $\Delta\hat{\rho}_1 = -0.014$, $z = -0.21$: an unchanged ρ_1 is not rejected
- **Minimum detectable effect (MDE)**: the smallest true change found with 80% power at the 5% level
 - ▶ $\text{MDE} = (z_{0.975} + z_{0.80}) \times \text{SE}(\Delta\hat{\rho}_1) \approx 2.8 \times \text{SE} = 0.18$
 - ▶ if Bitcoin was already close to efficient before 2024, this design could not detect a smaller improvement
- Placebo at 11 January 2016–2021: 0 of 6 significant, largest $|z| = 1.57$
- Causality would also need a control asset (Ethereum before its own ETFs in July 2024), parallel pre-trends and no spillovers

 ai_discovery_case.py

Mini-Case: Estimate, Placebo and MDE



- Left: $\hat{\rho}_1$ before and after, with robust 95% intervals; the difference with $\pm$ MDE
- Right: the ETF z lies inside the range produced by six placebo dates
- "Not significant" is not "no effect": the MDE is several times the estimate
- A better outcome: intraday price discovery across venues, not daily ρ_1

ai_discovery_case.py

Your Turn: A Project Seed

- ▣ **Question:** did the spot ETFs change the time-varying efficiency of Bitcoin, relative to a control asset?
 - ▶ rolling VR and DFA (this chapter) for Bitcoin and Ethereum; Ethereum is untreated until July 2024
 - ▶ if parallel trends, no anticipation and no spillovers are not credible, report descriptive, not causal, differences
- ▣ **Replicate first:** the chapter's yearly Bitcoin autocorrelations and the mini-case numbers
- ▣ **Pre-register:** H_0 , H_1 , dates, windows and corrections, committed before the results
- ▣ **Report:** robust inference, placebo tests and the MDE; the code as a Quantlet
- ▣ **Log:** prompts in `AI_USE.md`; every caught error (an i.i.d. standard error, an invented reference) in `AI_ERRORS.md`

Synthesis, Check Yourself, and Next: Chapter 3

Five steps for every efficiency claim

- ▣ **Information set:** which $\mathcal{I}_t$ (past prices, public news)?
- ▣ **Null:** RW1, RW3 or the martingale difference, and which risk model?
- ▣ **Statistic:** its variance valid under volatility clustering
- ▣ **Uncertainty:** horizons, markets and windows tried; out-of-sample evidence
- ▣ **Economics:** rejection $\neq$ exploitable profit $\neq$ irrational market

Check yourself

- ▣ Why can the S&P 500 reject RW3 and still be efficient in the economic sense?
- ▣ Why does BET's $H(20) > 0.5$ not prove long memory?
- ▣ Compare: which of BET's persistence and Bitcoin's non-rejection would survive a change of test, window and inference?
- ▣ Next, [Chapter 3](#): risk-adjusted returns with factor models, and the factor zoo

Appendix: Pioneers, Bachelier and Samuelson



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; [Wikimedia Commons](#)



Paul A. Samuelson

Photo: Bernard Gotfryd (1970–1975); public domain; [Wikimedia Commons](#)

- **Bachelier (1900)**: prices of French government bonds move like a random walk
 - ▶ “The mathematical expectation of the speculator is zero”
- **Samuelson (1965)**: if prices are properly anticipated, their changes are unpredictable
 - ▶ Unpredictability is a sign of a well-functioning market, not of chaos
- Both results are the ancestors of the Efficient Market Hypothesis



Appendix: Eugene Fama



Eugene F. Fama, Nobel Prize 2013

Photo: [Bengt Nyman \(2013\)](#); CC BY 2.0; Wikimedia

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- [Fama \(1970\)](#) gave the definition used in this chapter and the three forms of efficiency
- Nobel Prize 2013, shared with Robert Shiller (a sceptic of efficiency) and Lars Peter Hansen
- Update and survey: [Fama \(1991\)](#); accessible critique: [Malkiel \(2003\)](#)

Appendix: Data Inventory and Cleaning Sensitivity

Group	Series	Period
Developed equity	S&P 500, Euro Stoxx 50, Nikkei 225, Hang Seng	2000–2026
Emerging/frontier equity	BET, WIG20, BUX, PX, BIST 100, Bovespa, IPC, Nifty 50, Shanghai	2000–2026
Crypto-assets	Bitcoin, Ethereum	2014/2016–2026
Exchange rate	EUR/RON, BNR reference rate	2005–2026

- Daily closing values up to 18 September 2026; sample sizes T by market: summary table on the next slide
- Removing closes equal to the previous close also drops the rare genuine zero-return day (S&P 500: 3 of about 6 700 days); this moves the test statistics by less than 0.05
- Univariate tests use each series on its own calendar; multi-series analyses align prices on common trading days before computing returns

Appendix: Summary Table, Weak-Form Tests 2000–2026

Market	T	$\hat{\rho}_1$	robust SE	$\widehat{VR}(5)$	$z^*(5)$	CD p	Runs z
S&P 500 (US)	6 714	−0.099	0.027	0.830	−2.77	0.001	4.34
Euro Stoxx 50	6 836	−0.029	0.019	0.910	−2.08	0.097	3.41
Nikkei 225 (Japan)	6 545	−0.038	0.024	0.923	−1.48	0.317	1.97
Hang Seng (HK)	6 583	−0.004	0.024	0.967	−0.64	0.872	−0.17
BET (Romania)	6 670	0.109	0.028	1.173	2.98	0.001	−6.05
WIG20 (Poland)	6 685	0.018	0.018	1.009	0.24	0.777	2.25
BUX (Hungary)	6 666	0.037	0.024	1.013	0.26	0.427	−0.42
PX (Czechia)	6 682	0.052	0.032	1.041	0.59	0.350	−2.53
BIST 100 (Turkey)	6 692	0.005	0.024	1.033	0.64	0.949	−0.63
Bovespa (Brazil)	6 622	−0.028	0.025	0.928	−1.29	0.521	0.88
IPC (Mexico)	6 689	0.070	0.019	1.039	0.93	0.001	−3.27
Nifty 50 (India)	6 606	0.048	0.024	1.048	0.91	0.178	−5.46
Shanghai (China)	6 474	0.021	0.019	1.046	1.11	0.409	1.02
Bitcoin	4 384	−0.022	0.026	0.991	−0.19	0.835	2.68
Ethereum	3 913	−0.012	0.027	1.067	1.17	0.027	3.91
EUR/RON (reference rate)	5 352	0.170	0.036	1.157	2.19	0.000	−1.66

□ T : number of daily returns; CD: Chow–Denning over $q = 2, 5, 10, 20$; EUR/RON from July 2005, crypto on 365 days per year

Appendix: Runs Test on All Markets, Other VR Tests

- ▣ Runs z (table on the previous slide): Nifty 50 -5.46 , IPC -3.27 (persistence in emerging markets); Ethereum 3.91 , significant although $\hat{\rho}_1$ is not
- ▣ Choi (1999): an automatic VR test that chooses the horizon from the data (spectral estimate at frequency zero)
- ▣ Wild-bootstrap versions for small samples: Kim (2006); applications to Asian markets: Kim & Shamsuddin (2008)

Appendix: Automatic Portmanteau Test of the MDH

- Escanciano & Lobato (2009): $AQ = T \sum_{j \leq \tilde{p}} \hat{\gamma}_j^2 / \hat{\kappa}_j \rightarrow \chi_1^2$, with $\hat{\gamma}_j = \frac{1}{T-j} \sum_t e_t e_{t-j}$ and $\hat{\kappa}_j = \frac{1}{T-j} \sum_t e_t^2 e_{t-j}^2$ (robust to conditional heteroskedasticity)
 - ▶ $\tilde{p} \leq 10$ maximises $Q_p = T \sum_{j \leq p} \hat{\gamma}_j^2 / \hat{\kappa}_j$ minus a penalty that switches from BIC ($p \ln T$) to AIC ($2p$) when some $\sqrt{T}|\hat{\rho}_j|$ is large
- AQ: S&P 500 13.1 ($p = 0.0003$), BET 14.9 ($p = 0.0001$), Bitcoin 0.8 ($p = 0.385$); 5 of 16 series reject at 5%

Appendix: Predictive Regression, Data and Full Output

- Monthly log excess return (dividends included, minus the 3-month T-bill rate) on lagged dp_t , $T = 1111$ months
 - ▶ Data: [Shiller \(2026\)](#) (S&P Composite, to 2026-06), extended with our S&P 500 closes and the dividends of the SPY fund; 3-month US Treasury bill yield (FRED, series TB3MS; annual %), monthly log rate $\ln(1 + \text{TB3MS}/1200)$ lagged one month
- OLS: $\hat{\beta} = 0.267$, Newey–West SE 0.293 (12 lags), $t = 0.91$, OLS $t = 1.16$, $R^2 = 0.12\%$
- Corrected slope / unadjusted HAC SE = -0.22 : only illustrative, it ignores the uncertainty of the correction
- Economic value [Campbell & Thompson \(2008\)](#): the expected return of a mean–variance investor who times the market with the forecast rises by the proportion
 - ▶ $\frac{R^2}{1 - R^2} \cdot \frac{1 + S^2}{S^2}$, S the monthly Sharpe ratio; with $S = 0.160$ and $R^2 = 0.5\%$: 20%
- Diebold–Mariano $t = -0.57$ for the same forecasts, against Clark–West $t = 0.42$

Appendix: Harold Hurst and the Nile



Harold Edwin Hurst (1880–1978)

Photo: Elliott & Fry (1953); public domain; Wikimedia

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- Hurst (1951) designed Nile reservoirs: wet years followed wet years more often than chance allows, measured with the rescaled range



Nilometer on Elephantine Island, Aswan

Photo: Olaf Tausch (2011); CC BY 3.0; Wikimedia Commons

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Appendix: The Anis–Lloyd Expectation

- ▣ Even for i.i.d. returns, the slope of $\ln(R/S)$ on $\ln n$ exceeds 0.5 for small n
- ▣ Anis & Lloyd (1976): exact $\mathbb{E}[R_n/s_n]$ under i.i.d. Gaussian returns
 - ▶ $\mathbb{E}[R_n/s_n] = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma(n/2)} \sum_{i=1}^{n-1} \sqrt{\frac{n-i}{i}}$ (Γ : the gamma function)
- ▣ Corrected exponent: $H_{AL} = 0.5 + \text{slope of } \ln(R/S) - \ln \mathbb{E}[R_n/s_n] \text{ on } \ln n$
- ▣ Uncorrected estimates overstate memory by about 0.04–0.05 with daily data

Appendix: Long Memory with Inference, the Local Whittle Estimator

- $d = H - 0.5$; spectral density $f(\lambda) \sim G\lambda^{-2d}$ as $\lambda \rightarrow 0$; fit only the first m Fourier frequencies $\lambda_j = 2\pi j/T$
 - ▶ $\hat{d} = \arg \min_d \left[\ln \left(\frac{1}{m} \sum_{j=1}^m \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^m \ln \lambda_j \right]$, I : the periodogram
 - ▶ Robinson (1995): for $|d| < 1/2$, $m \rightarrow \infty$, $m/T \rightarrow 0$, a smooth f near 0 and a linear process with conditionally homoskedastic martingale-difference innovations, $\sqrt{m}(\hat{d} - d) \rightarrow N(0, 1/4)$ without Gaussianity: a confidence interval instead of a band
- $m = T^{0.65}$, 95% CI $\hat{d} \pm 1.96/(2\sqrt{m})$:
 - ▶ S&P 500 -0.013 $[-0.069, 0.043]$; Bitcoin 0.063 $[-0.002, 0.127]$
 - ▶ BET 0.118 $[0.062, 0.174]$; Ethereum 0.112 $[0.045, 0.179]$
- Bandwidth trade-off: a large m reaches frequencies where short-run autocorrelation (BET $\hat{\rho}_1 = 0.109$) biases $\hat{d}$ upward; with $m = T^{0.5}$ BET still gives 0.135 (SE 0.056): a sensitivity check supporting, not proving, that its $d > 0$ is not only short memory (GARCH returns also strain the conditions above)

Appendix: EMH versus AMH

	EMH	AMH
Investors	Rational, optimising	Bounded rationality, heuristics
Efficiency	An equilibrium property, tested as a restriction	A degree that changes with conditions
Risk–return	Equilibrium risk premia, possibly time-varying	Changes with the population of investors
Anomalies	Mispriced risk or data mining	Transient profit opportunities
Empirical tool	Tests of an equilibrium restriction	Tests of time and state variation (rolling windows)

Appendix: Is Bitcoin Efficient? The Literature

- [Urquhart \(2016\)](#): Bitcoin inefficient over 2010–2016; some tests indicate efficiency in the later subperiod
 - ▶ His battery (Ljung–Box, runs, Bartels, automatic VR, BDS, R/S) includes BDS, which rejects under volatility clustering alone: not a test of RW3
- [Nadarajah & Chu \(2017\)](#): with a power transformation of returns, Bitcoin is weak-form efficient
- [Kristoufek \(2018\)](#): Bitcoin markets mostly inefficient in 2010–2017, with efficiency changing over time
- [Tran & Leirvik \(2020\)](#): time-varying efficiency across many crypto-assets, increasing over time
- [Pele et al. \(2023\)](#): crypto-assets have become a distinct asset class; their returns share the stylised facts of other markets
- Year by year in this chapter: Bitcoin 2011–2013 $\hat{\rho}_1 = -0.002, -0.071, 0.069$; the largest $|z^*(5)|$ in any year is 1.71

Appendix: Romania, Frontier then Emerging

- ▣ [FTSE Russell \(2019\)](#): Romania reclassified from Frontier to Secondary Emerging, announced in September 2019, effective 21 September 2020
- ▣ Expected channels: more foreign investors, more analysts, more liquidity $\Rightarrow$ faster price discovery
- ▣ Earlier BVB evidence: crashes and log-periodic bubbles [Pele & Mazurencu-Marinescu \(2012\)](#); herding in an illiquid market [Pele, Mazurencu-Marinescu & Nijkamp \(2013\)](#)
- ▣ Did efficiency improve? Seminar C1 compares five years before and after, with WIG20 and BUX as a comparison group and a block bootstrap
 - ▶ Block bootstrap: resamples blocks of consecutive days to keep the dependence [Künsch \(1989\)](#); a before/after comparison is descriptive unless confounders are controlled

Appendix: How Fast Is “Fast”?

- At high frequency, futures prices react to scheduled macro news within microseconds; [Tak, Schlamp, Osterrieder & Pele \(2026\)](#) measure this race at Eurex and CME
- Crypto events: [Pele, Lin, Wang & Gaman \(2025\)](#) study market responses to Ethereum development milestones
- Speed of adjustment is itself a measure of efficiency, and it depends on technology ([Chapter 10](#), market microstructure)

Appendix: Dependent Tests, Reality Check, SPA and StepM

- Bonferroni and Holm ignore the correlation between tests (same markets, same days)
 - ▶ [White \(2000\)](#): Reality Check, bootstrap of $\max_j$ over all rules tried; [Hansen \(2005\)](#): SPA, studentised and less sensitive to poor rules
 - ▶ [Romano & Wolf \(2005\)](#): StepM (Romano–Wolf step-down procedure), step-down control of the FWER from the **joint** bootstrap distribution of all statistics
- Our 15 calendar tests: StepM with a bootstrap of blocks of 3 calendar months, common to all markets, 999 replications
 - ▶ Critical p at the first step 0.0016 (Bonferroni: 0.0033); 2 rejections, BET and BUX TOM, as with Holm
 - ▶ Why stricter than Bonferroni? At nominal 5%, the bootstrap rejection rates of the HAC tests are 6.0% (day of week), 6.4% (TOM), 6.4% (January)
- From over two million simulated strategies: thresholds $|t| > 3.8$ (time series) and 3.4 (cross-section); ignoring them makes about 45% of rejections false [Chordia, Goyal & Saretto \(2020\)](#); lowering false discoveries raises missed ones [Harvey & Liu \(2020\)](#)

Appendix: Time-Series Momentum, All Markets

Market	Mean (%/month)	t_{HAC}	Sharpe TSMOM	Sharpe buy-and-hold	$\hat{\rho}_1$ monthly
S&P 500 (US)	0.61	2.22	0.48	0.52	0.03
Euro Stoxx 50	0.16	0.51	0.11	0.16	0.05
Nikkei 225 (Japan)	0.61	1.85	0.39	0.42	0.09
Hang Seng (HK)	0.18	0.53	0.10	0.19	0.04
BET (Romania)	1.40	2.68	0.65	0.75	0.19
WIG20 (Poland)	0.67	1.86	0.36	0.26	0.02
BUX (Hungary)	0.72	1.79	0.40	0.64	0.10
PX (Czechia)	0.38	0.99	0.24	0.45	0.10
BIST 100 (Turkey)	1.03	1.85	0.37	0.75	-0.08
Bovespa (Brazil)	0.28	0.65	0.14	0.51	0.11
IPC (Mexico)	0.44	1.41	0.32	0.63	0.03
Nifty 50 (India)	0.50	1.29	0.28	0.64	0.02
Shanghai (China)	0.59	1.32	0.30	0.23	0.08
Bitcoin	4.47	1.94	0.72	1.08	0.13
Ethereum	6.50	1.28	0.57	0.90	0.16

- 307 months for equities, fewer for crypto; Sharpe ratio $\sqrt{12} \bar{x}/s_x$ of price returns, no risk-free rate subtracted; Moskowitz, Ooi & Pedersen (2012) use futures excess returns scaled to 40% ex-ante volatility

Appendix: Robert Shiller and Richard Thaler



Robert J. Shiller, Nobel Prize 2013

Photo: Bengt Nyman (2013); CC BY 2.0; Wikimedia Commons

- Shiller (1981): stock prices move far more than the dividends that should justify them (excess volatility)
- De Bondt & Thaler (1985): investors overreact; extreme losers become winners over the next 3–5 years



Richard H. Thaler, Nobel Prize 2017

Photo: Bengt Nyman (2017); CC BY 2.0; Wikimedia Commons



Appendix: Case Study, Research Question

- [Martin & Nagel \(2022\)](#): investors price N assets whose cash flows depend on J characteristics with unknown coefficients; what does a rejected no-predictability test mean when J is comparable to N ?
 - ▶ the coefficients must be learned from past data: a high-dimensional prediction problem, as in machine learning
- Semi-strong efficiency: prices “fully reflect” public information [Fama \(1970\)](#); two readings
 - ▶ investors know the parameters of the cash-flow model (rational expectations: the null of most efficiency tests)
 - ▶ investors update their beliefs optimally (Bayes) with the data available in real time
- Why it matters: hundreds of in-sample return predictors, the “factor zoo” [Harvey, Liu & Zhu \(2016\)](#); text, satellite images and social media keep increasing J
 - ▶ does every significant predictor need a risk-based or a behavioural explanation?
- Link to this chapter: in-sample vs out-of-sample predictability [Campbell & Thompson \(2008\)](#), [Goyal, Welch & Zafirov \(2024\)](#), and the decay of anomalies after publication [McLean & Pontiff \(2016\)](#)
- Investors are risk-neutral in the model: no joint-hypothesis problem, so no predictability can be a risk premium



Appendix: Case Study, Model and Method

- **Model** (Martin & Nagel (2022), Sec. 2; section, equation, proposition and figure numbers in this appendix refer to the paper): cash-flow growth $\Delta y_t = Xg + e_t$, X the $N \times J$ matrix of characteristics, $e_t \sim N(0, I)$; zero interest rate
 - ▶ prior $g \sim N(0, (\theta/J)I)$, the true distribution from which g is drawn: the posterior mean is a ridge regression (OLS with a penalty on the squared coefficients), i.e. coefficients shrunk towards 0 (Prop. 1)
 - ▶ realised return = underreaction caused by shrinkage + estimation noise + unpredictable shock (Prop. 2)
- **Tests**: in sample, a cross-sectional OLS of r_{t+1} on X (requires $J < N$) with a joint χ^2 test; out of sample, a portfolio with weights $\frac{1}{N}Xh_{s+1}$ from coefficients estimated in other periods
 - ▶ $r_{\text{OOS},t+1} = \frac{1}{N}h'_{s+1}X'r_{t+1}$, $s \neq t$ (eq. 19); r_{15} : the same with $s = t$
- **Asymptotics and simulations**: $N, J \rightarrow \infty$ with $J/N \rightarrow \psi > 0$; simulations with $N = 1000$, entries of X drawn from the standard Normal distribution and $\theta = 1$ (Sec. 4)
- **Empirical design** (Sec. 7): US stocks in CRSP, without those below the 20th NYSE size percentile or priced below \$1
 - ▶ predictors: simple and squared monthly returns at lags $t - 2$ to $t - 120$ (238 predictors), demeaned and standardised across stocks every month
 - ▶ full-sample ridge, 1971 – June 2019, penalty chosen by leave-one-year-out cross-validation; rolling 20-year OLS gives r_{OOS} from January 1956

Appendix: Case Study, Formal Results

- In sample (numbering as in [Martin & Nagel \(2022\)](#)): the rejection probability of the no-predictability test tends to 1 as $N, J \rightarrow \infty$ (Prop. 4); for $\psi > 0.4$ ($\theta = 1, t = 1$) the probability of no rejection is of the order $e^{-0.015N}$ (Fig. 2)
 - ▶ with fully rational Bayesian investors, the rational-expectations null is almost surely rejected: a rejection is not evidence of irrationality
- Simulations ($N = 1000$, four periods of learning): a test of nominal size 5% rejects with probability above 90% once $J > N/2$ (Fig. 3b)
 - ▶ not a size distortion: the in-sample predictable component really is in the prices
- Single-predictor tests with k -FWER control (probability of k or more false rejections at most 5%): for $k = 20$ and $J/N = 0.05$, the probability of at least k rejections is 25% instead of 5%; about 80% for J/N close to 1 (Fig. 4)
 - ▶ multiple-testing corrections do not help: the problem is the null hypothesis, not the test
- Out of sample: $E[r_{\text{OOS}, t+1}] = 0$, forward and backward in time (Prop. 6); positive only if investors shrink more than the data warrant (Fig. 6)
 - ▶ out-of-sample tests keep their economic meaning
- US stocks: the ridge coefficients recover momentum, long-run reversal and momentum seasonality (Fig. 7); $r_{15} \approx 0.006\%$ per month, an in-sample R^2 of about 1%, stable over time (Fig. 8)
 - ▶ r_{OOS} is much lower and often near 0; in the 10-year windows ending in 2014–2019 it is less than two standard errors above 0: little of the in-sample predictability survives

Appendix: Case Study, Replication, Limitations and Extensions

- **Replicate with the course data** (stock-level CRSP files are not part of them)
 - ▶ 49 industry portfolios, monthly value-weighted returns from July 1926 ([Kenneth French Data Library](#)); the same 238 predictors, monthly standardisation and rolling 20-year windows (Seminar 2, C3)
 - ▶ data limitation: 49 portfolios instead of thousands of stocks; the size and price filters do not apply to portfolios
- **Limits stated by the authors**
 - ▶ lagged returns are equilibrium outcomes, not the exogenous X of the model: the application illustrates the wedge, it does not test the model
 - ▶ a t -test on the monthly r_{OOS} is not a valid test of $E[r_{\text{OOS}}] = 0$: the expectation also averages over g
 - ▶ the backward result may rely on i.i.d. cash-flow growth; homogeneous investors and constant characteristics
- **Open extensions**
 - ▶ time series, as the authors suggest: does the dividend–price regression of this chapter show the same in-sample/out-of-sample wedge?
 - ▶ nonlinear models (trees, neural networks [Gu, Kelly & Xiu \(2020\)](#)) add functional flexibility without necessarily changing J : does the wedge grow?
 - ▶ AMH: is early out-of-sample predictability the excess shrinkage of investors with less computing power?

Appendix: Case Study, In Sample vs Out of Sample in Detail

- r_{IS} is positive in every 20-year window, as the in-sample OLS fit guarantees ($r_{IS} = b'G^{-1}b/240 \geq 0$, $G = \sum_s X_s'X_s/N_s$ and $b = \sum_s X_s'r_s/N_s$ over the window): from 5.90 to 15.14 ($\times 10^{-5}$ per month), an average in-sample R^2 of 5.4%
 - ▶ about 1% for individual stocks in [Martin & Nagel \(2022\)](#): 238 predictors against only 40–49 portfolios a month, pooled over 240 months, inflate the in-sample fit
- Out of sample, July 1956 – August 2026 (842 months): mean $r_{OOS} = 1.05 \times 10^{-5}$, about 11% of the mean r_{IS}
- Until June 1986: mean $r_{OOS} = 1.98 \times 10^{-5}$, 4.16 standard errors above 0; afterwards 0.36×10^{-5} , only 0.62 standard errors
 - ▶ the 10-year average is last above two standard errors in the window ending in December 1987; in the windows ending in 2014–2019 its largest ratio to the standard error is 1.25, as for stocks
- Interpretation suggested by [Martin & Nagel \(2022\)](#): the gap is consistent with investor learning, but does not establish it or rule out risk premia; early out-of-sample gains are also consistent with excess shrinkage before the computer era (Fig. 6)
- The bands are descriptive: a t -test on the monthly r_{OOS} is not a valid test of $E[r_{OOS}] = 0$

Appendix: The Discovery Loop, Reusable Prompts (1/2)

□ Literature

- ▶ List peer-reviewed studies since 2020 on how ETF introductions change the efficiency or price discovery of the underlying. For each: DOI, design (event study, difference-in-differences (DiD), structural break), efficiency measure. Flag any entry you are unsure exists.
- ▶ Resolve every DOI on Crossref; the title and authors must match

□ Hypothesis

- ▶ Give three mechanisms by which spot ETFs could change Bitcoin's serial dependence (arbitrage, investor mix, trading hours). For each, the sign of the change in ρ_1 and $VR(q)$, and a variable that separates it from the others.

□ Data and code

- ▶ Read the daily Bitcoin closing prices; log returns; ρ_1 with a heteroskedasticity-robust standard error and the Lo-MacKinlay $VR(5)$ with z^* , in two windows of equal length around 11 January 2024.
- ▶ Replicate a known number first: the chapter's full-sample Bitcoin $\hat{\rho}_1 = -0.022$ (robust SE 0.026)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

▣ Robustness

- ▶ Write a pre-registered robustness plan: window lengths, weekday-only closes, $VR(q)$ for $q = 2$ to 20 with Chow-Denning, placebo dates, and Ethereum as a control until its own spot ETFs (July 2024).

▣ Critique

- ▶ Act as a hostile referee: if Bitcoin was already efficient before 2024, what could this test ever detect? Compute the minimum detectable change in ρ_{01} and name a better outcome variable.

▣ Report

- ▶ Estimates, robust standard errors, the placebo distribution, every test run (with a multiple-testing correction)
- ▶ Prompts that shaped a decision go into AI_USE.md

- ▣ The assistant drafts the tests; the human decides which one answers the question

Appendix: What the Researcher Must Check

- **i.i.d. standard errors:** $1/\sqrt{T}$ rejects too often under volatility clustering
 - ▶ test: robust SE and z^* , as in the weak-form tests of this chapter
- **Unequal windows:** different power before and after; test: equal lengths, fixed in advance
- **Event date chosen from the data**
 - ▶ test: the regulatory date only, plus a placebo distribution at dates without an event
- **Confounding:** before/after is not causal; test: a control asset (Ethereum before July 2024) in a difference-in-differences
 - ▶ causal only with parallel pre-trends, no anticipation and no spillover from Bitcoin ETFs to Ethereum; the common post-period (Jan–Jul 2024) is short; otherwise the comparison is descriptive
- **Many statistics:** ρ_1 , VR(2–20), runs; test: Chow–Denning or Holm, not the best single p -value
- **Low power:** report the minimum detectable change, ≈ 2.8 standard errors of the difference

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

-  Efficiency of cryptocurrency markets - A GMM-based analysis
-  Generalized Method of Moments

Code, data and textbooks

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_02
- Textbooks: Campbell, Lo & MacKinlay (1997), Ch. 2; Franke, Härdle & Hafner (2019)

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